

GRADES 7-10



EMERGENCY SUB PLANS PACK

for **DIGITAL TECHNOLOGIES**

★ Keep learning on track—even when plans change. ★

VOLUME 4

**COMPUTER
SYSTEMS**



CPU

RAM

STORAGE

**READY
WHEN YOU
NEED IT!**

TASK CARD

12

Write a program that
flashes an LED on
for 2 seconds and
off for 2 seconds.



NETWORKS

SHORT ANSWER QUESTIONS



1. What is the main purpose of a router?

2. What is an IP address?

3. Give one example of a wireless network.

BINARY BASICS

Convert the following
binary numbers to decimal.

1010	=	_____
1101	=	_____
1111	=	_____
10011	=	_____



**NO-PREP
ACTIVITIES**



**READY-TO-USE
RESOURCES**



**PERFECT FOR
EMERGENCY
LESSONS**



**GREAT FOR
SUBSTITUTE
TEACHERS**



**ENGAGING
DIGITAL TECH
TASKS**



**PRINT OR
DIGITAL USE**



YOUR **LIFESAVER** IN THE CLASSROOM

⌚ SAVE TIME

⊗ REDUCE STRESS

👥 KEEP STUDENTS
ON TRACK



**MADE BY TEACHERS
FOR TEACHERS**

Set this up while you are well

15 MINUTES

You are reading this because you are **well**. Spend fifteen minutes now and this pack will run itself on a day when you are not.

The one job you have today

Print pages 3–6 of this pack and leave them in a clearly labeled folder where your substitute teacher will find them — top drawer, on the desk, or with the front office. Everything a substitute needs is in those four pages.

You do not need to print the lessons in advance. The substitute teacher chooses one on the day and photocopies it.

SETTING IT UP · FIFTEEN MINUTES, ONCE

- 1 **Print the whole pack once.** Put it in a display folder or bind it. Write your name and room number on the cover.
- 2 **Print pages 3–6 a second time** and clip them to the front. These are the pages a substitute teacher reads first.
- 3 **Tell your department head where it lives.** On a genuine emergency day you will not be answering your phone, so somebody else needs to know.
- 4 **Cross off the lessons you have already used.** There is a tracker on page 4. Two years of substitute days will not repeat if you keep it current.
- 5 **Optional: leave a class list** and a note about any students who need specific support, seating or a reader.

Why these lessons are unplugged

On a substitute day the laptop cart is often locked, the login server is down, or nobody knows the room password. Every lesson here runs on paper.

Each lesson also has an **If the computers are working** note, so the same sheet still works when technology cooperates.

Why they are not busy work

Every lesson targets a genuine Digital Technologies concept and produces marked evidence you can put in a folder.

If you are away for a week, running four of these in sequence is a defensible unit of work, not a holding pattern.

A word about the answer keys

Every answer key sits on the **last page of its lesson**. If you hand a lesson to a substitute teacher, hand them the answer key too — it is the single thing that lets a non-specialist mark with confidence and answer a student who says “*but why?*”

This is Volume Four

Volume 4 stands completely on its own and shares nothing with Volumes 1, 2 or 3. There is no binary conversion, no flowchart, no data cleaning, no truth table and no DNS here. Owning all four gives you forty substitute lessons that never repeat.

You do not need to know this subject

You have walked into a Digital Technologies class. You may have never taught the subject. That is completely fine — this pack was written for you.

Read this and you are ready

You do not need to know how to code. Every lesson is a paper worksheet with a full answer key. Your job is to hand it out, keep the room settled, and check work as students finish. You are not expected to teach the content.

DO THESE FOUR THINGS

- 1 **Pick one lesson** from the table on the next page. Any of them works for any class from Grade 7 to Grade 10. If you cannot decide, use Lesson 04.
- 2 **Photocopy the student pages only.** Each lesson tells you exactly which page numbers to copy. One copy per student.
- 3 **Keep the teacher page and the answer key.** Do not photocopy those.
- 4 **Read the teacher page once** before the bell. It takes about two minutes and includes the exact words you can say to start the lesson.

If a student says “we have done this”

Ask them to prove it by completing the extension section at the end. If they genuinely have, switch them to a different lesson from the table — the pack has ten, and they will not have done them all.

If you finish early

Every lesson has an extension task on its final student page. Point students there. There is also a discussion prompt on the teacher page you can run as a whole class for the last five minutes.

What to leave for the regular teacher

Collect the completed sheets. Fill in the handover note on the last page of this pack — it takes ninety seconds and tells the regular teacher which lesson you used, how far the class got, and anything that happened.

Leave both on the desk.

Every lesson is standalone. There is no order and no prerequisite — pick whichever suits the class in front of you.

#	LESSON	WHAT IT COVERS	DIFFICULTY	MINS	STUDENT PAGES
01	Pictures Made of Numbers	Bitmaps, resolution and color depth	★★	60	7-9
02	Sound Made of Numbers	Sampling, bit depth and audio files	★★	55	12-14
03	Stacks, Queues and the Undo Button	Data structures in real systems	★★★	55	17-19
04	Reading the Logs	Digital forensics and evidence	★★★	60	22-24
05	Version Control by Hand	Tracking changes and collaboration	★★	55	27-29
06	Building a School Network	Network devices, topology and capacity	★★	60	32-34
07	Design for Everyone	Accessibility and inclusive design	★	55	37-39
08	The Algorithm Decided	Automated decisions and fairness	★★	55	42-44
09	The Smart Home Audit	Connected devices and their risks	★★	55	47-49
10	Will a Machine Take This Job?	Automation, work and the future	★	55	52-54

Easiest to supervise

07, 09 and 10. Judgment and argument with a definite conclusion waiting at the end. Students have opinions, so nobody gets stuck.

Best for a quiet, working class

01, 02 and 03. Puzzle-like with exact answers. Strong students will race; that is what the extensions are for.

Best for a restless class

04, 05 and 08. An investigation, a folder full of chaos and a scholarship to argue about. Noisier, but productively so.

Used-lesson tracker · for the regular teacher

LESSON	DATE USED	CLASS	NOTES
01 — Images as data			
02 — Sound as data			
03 — Stacks and queues			
04 — Digital forensics			
05 — Version control			
06 — Network design			
07 — Accessibility			
08 — Automated decisions			
09 — Smart devices			
10 — Automation and work			

The first ten minutes

SCRIPT

The first four minutes decide the next fifty. Here is a script that works even if you have never met this class.

SAY THIS AT THE DOOR

"Come in, sit down, and leave your bags off the desks. You will not need a computer today. Everything is on paper."

THEN, ONCE THEY ARE SEATED

"Your teacher is away. My name is _____. We are doing a proper Digital Technologies lesson today, not a free period. There is a worksheet, it gets collected at the end, and your teacher will see it."

Why that last sentence matters

The single most useful thing you can say is that the work will be seen by their regular teacher. It is also true — there is a handover note at the back of this pack.

THE SHAPE OF EVERY LESSON IN THIS PACK

TIME	WHAT HAPPENS	WHAT YOU DO
0-5	Settle, hand out sheets	Use the script above
5-10	Read the opening page together	Ask one student to read it aloud
10-45	Students work through the tasks	Circulate. Check answers against the key
45-55	Extension task for those who finish	Point them to the last student page
55-60	Collect sheets, pack up	Fill in the handover note

If a student refuses to work

Offer the smaller ask: *"Just do question one and show me."* Most refusals are about not knowing where to start, not defiance.

Note it on the handover sheet. Do not escalate a single lesson into a battle you will not be there to finish.

If the whole class claims they cannot do it

Do the first question together on the board. Every lesson has a worked example on the first student page.

If it is genuinely too hard for the grade level, switch to Lesson 07 or Lesson 10 — both need no prior knowledge at all.

Photocopying reminder

Student pages only. Each lesson's teacher page states the exact page range. The **answer key is always the last page** of the lesson — keep it.

Pictures Made of Numbers

2 MINUTES

Students decode an image from its binary, encode one of their own, then calculate what a photograph actually costs in bytes and work out why enlarging one turns it into squares.

You do not need to know this subject

Every answer is a shaded grid or a multiplication. The completed images and every number are in the key.

Photocopy these pages only

Pages 7-9

One per student. **Do not photocopy this page or the answer key.**

What students need

A pen, and a pencil for shading squares.

WHAT TO SAY TO START

"A photograph contains no faces, no colors and no sky. It contains several million numbers. Today you find out exactly which numbers, and how many."

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0-5	Hand out sheets, settle	Use the door script on page 5
5-14	Read the pixel panel and the worked row	Shade the worked row together
14-28	Task 1.1 — decode and encode	Self-checking: a shape appears or it does not
28-42	Task 1.2 — color depth and file size	Arithmetic. Every answer is in the key
42-54	Task 1.3 — resolution, and why zooming fails	The vector question is the payoff
54-60	Tasks 1.4 and 1.5, collect sheets	Discussion prompt below

Questions students will ask

"Is 1 black or white?"

It is whatever both ends agreed. On this sheet 1 means shaded. Somebody has to decide, and that decision is part of the format.

"Why 24 bits for color?"

Eight bits each for red, green and blue. Three numbers per pixel, each from 0 to 255.

"My file size looks wrong."

Check they multiplied the pixels first, then the bits per pixel, then divided by 8 to get bytes.

If they finish early

Task 1.4 covers metadata. Strong students can then work out why an image posted publicly can reveal where it was taken, and what a platform does about that.

If the computers are working

Students can open any photo's properties and compare the stored file size with the number they calculated. The stored file is far smaller, which is a neat lead-in to why compression exists.

Curriculum focus · for the regular teacher's records

How digital systems represent images; pixels, resolution and color depth; calculating file sizes; bitmap and vector graphics; metadata. Discussion prompt for the last five minutes: *"The screen is showing a face. Where is the face actually stored, and in what form?"*

Every picture is a grid

16 MIN

NAME _____

CLASS _____

DATE _____

One bit per pixel

A black and white image is a grid of squares. Each square gets one bit: **1** means shaded, **0** means blank. Read left to right, top to bottom, and the picture rebuilds itself.

A color image works identically, except each pixel needs more bits, because it has to say which of millions of colors it is.

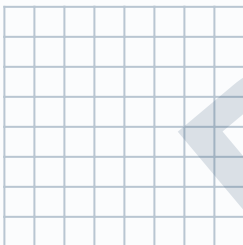
Task 1.1 · Decode, then encode

DO

a · decode this

00111100
01000010
10100101
10000001
10100101
10011001
01000010
00111100

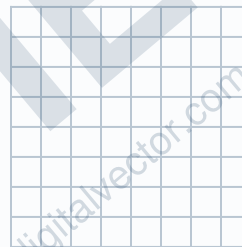
Shade every 1.



What is it? _____

b · now encode

Shade a capital **T** in this grid, then write the eight binary rows beside it.

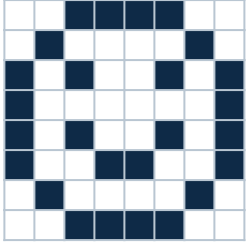


_____	_____
_____	_____
_____	_____
_____	_____

How many bits did your T take? _____

c. Both images are 8 by 8. Explain why they take exactly the same number of bits even though one is far more complicated. _____

Task 1.1 · the images



It is a face — a round outline, two eyes and a smile.

b. A capital T is 11111111 then rows of 00011000. Any recognizable T is correct, provided the binary matches the shading exactly.

Both take 64 bits — 8 rows of 8. **c.** A bitmap stores one value per pixel whatever the picture shows, so blank space costs the same as detail. That is the weakness of the format, and the reason compression exists.

Task 1.3 · resolution

a. $1920 \times 1080 = 2,073,600$ pixels.

b. The original has 40,000 pixels; the screen needs 2,073,600. Over **two million** pixels have to be invented.

c. It guesses, by copying or averaging neighboring pixels. Nothing new is discovered, which is why the result is blocky or smeared. The detail was never in the file.

d. A vector stores instructions — circle here, this radius, this color — so it is redrawn at whatever size is asked for and is always exactly sharp.

B: grid of pixel colors; blurry when enlarged. **V:** shapes and instructions; and the photograph is **B**, because no reasonable set of instructions describes a face.

Task 1.2 · colors and sizes

Colors: $1 \text{ bit} = 2 \cdot 2 \text{ bits} = 4 \cdot 4 \text{ bits} = 16 \cdot 8 \text{ bits} = 256 \cdot 24 \text{ bits} = 16,777,216$.

a. $10,000 \times 1 \div 8 = 1,250 \text{ bytes}$

b. $10,000 \times 8 \div 8 = 10,000 \text{ bytes}$

c. $10,000 \times 24 \div 8 = 30,000 \text{ bytes}$

d. $12,000,000 \times 24 \div 8 = 36,000,000 \text{ bytes}$, about 36 MB

e. The file was compressed. A JPEG discards detail the eye is poor at noticing, turning 36 MB into about 3 MB. Students who name the loss as well have the better answer.

Task 1.4 and 1.5

a. Sorting photos by date, finding where a holiday photo was taken, proving when a picture was made, or matching a photo to a place on a map.

b. The coordinates of their home, accurate to a few yards, plus the date, the time and the device — none of which they meant to publish.

c. Platforms strip it to reduce file size and to avoid publishing user locations by accident. A photographer objects because the metadata carries their name, their copyright notice and their settings, all of which vanish with it.

The last word: the face is stored as a long list of numbers — three per pixel — and it only becomes a face when a screen lights up in that pattern and a human brain interprets it. Nothing in the file resembles a face in any way.

Measuring a wave

14 MIN

NAME _____

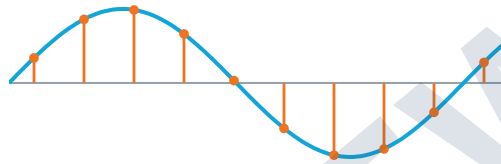
CLASS _____

DATE _____

Thousands of measurements a second

A microphone turns air pressure into a wave. The computer measures the height of that wave at regular instants and writes down a whole number each time.

Sample rate is how many measurements per second. **Bit depth** is how many different heights each measurement is allowed to be. Everything in between the measurements is gone forever.



Task 2.1 · Round the samples

CALCULATE

The wave was measured at ten instants. With **4 bits** each measurement may be any whole number from -8 to 7 . With **3 bits** only even numbers are available: $-8, -6, -4, -2, 0, 2, 4, 6$.

SAMPLE	1	2	3	4	5	6	7	8	9	10
True height	0.0	3.1	5.4	6.2	4.8	1.2	-2.6	-5.9	-6.4	-3.3
4-bit value										
3-bit value										

- Which sample loses the most accuracy at 3 bits? ____ How far off is it? ____
- The difference between the true height and the stored number has a name: quantisation error. What would a listener hear if it were large? ____
- Doubling the sample rate would change the table how? ____

The record nobody reads

16 MIN

NAME

CLASS

DATE

The incident

A school office server holds staff files. On Friday morning a payroll spreadsheet turned up on a public file-sharing site. Below is the server log for the previous day. Addresses beginning 10. are inside the school; anything else is outside it.

server.log · Thursday, August 14

AUDIT TRAIL

```
07:58 j.patel LOGIN 10.2.14.6 ok
08:31 j.patel OPEN /reports/term3.xlsx
09:02 a.novak LOGIN 10.2.14.9 ok
09:05 a.novak OPEN /staff/schedule.docx
12:44 j.patel LOGOUT
15:10 a.novak LOGOUT
18:55 MAILGW j.patel opened link in "Payroll update required"
19:12 j.patel LOGIN 203.0.113.44 ok
19:13 j.patel OPEN /admin/salaries.xlsx
19:15 j.patel DOWNLOAD /admin/salaries.xlsx
19:18 j.patel OPEN /admin/contracts/
19:22 j.patel DOWNLOAD /admin/staff_list.csv
19:26 j.patel LOGIN 203.0.113.44 FAILED (admin console) x3
19:31 j.patel LOGOUT
```

Task 4.1 · Read the evidence

INVESTIGATE

#	QUESTION	ANSWER
a	Which login is unusual, and what makes it unusual?	
b	How long did that session last?	
c	Which two files were taken?	
d	What was attempted at 19:26, and did it work?	
e	What happened 17 minutes before the unusual login?	

Photocopy this page or fill it in directly. Ninety seconds of your time saves the regular teacher a great deal of guesswork.

SUBSTITUTE TEACHER

CLASS

DATE

PERIOD

Lesson used

CHECK ONE

- | | |
|--|---|
| ☐ 01 · Pictures Made of Numbers | ☐ 06 · Building a School Network |
| ☐ 02 · Sound Made of Numbers | ☐ 07 · Design for Everyone |
| ☐ 03 · Stacks, Queues and the Undo Button | ☐ 08 · The Algorithm Decided |
| ☐ 04 · Reading the Logs | ☐ 09 · The Smart Home Audit |
| ☐ 05 · Version Control by Hand | ☐ 10 · Will a Machine Take This Job? |

How far did the class get?

- ☐ Everyone finished the main tasks
- ☐ Most finished; some are partway
- ☐ We got through about half
- ☐ We did not get far — see notes

Sheets

- ☐ Collected and left on the desk
- ☐ Students kept them
- ☐ Graded against the answer key

Anything the regular teacher should know

NOTES

Students to mention · for any reason, good or otherwise

NAMES

Thank you

Walking into an unfamiliar subject is not easy. Leaving a clear note behind is the part that makes the next teacher's day work, and it is genuinely appreciated.

Terms of use

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Curriculum alignment

CSTA K-12 Computer Science Standards, levels 2 and 3A. Across the ten lessons this volume touches:

- Representing images: pixels, resolution and color depth
- Representing sound: sampling, bit depth and file size
- Abstract data structures: stacks and queues
- Interpreting logs, evidence and audit trails
- Version control and collaborative development
- Network devices, topologies and capacity planning
- Accessibility and inclusive design
- Automated decision-making, fairness and accountability
- Security and privacy of connected devices
- Automation and the future of work

Individual lesson teacher pages name the specific focus for that lesson.

The rest of the series**Volume 1 · Foundations**

Algorithms, binary, trace tables, flowcharts, cyber attacks, networks, sorting, debugging, AI and digital footprints.

Volume 2 · Data, Systems and Design

Data cleaning, misleading charts, databases, input-process-output, compression, encryption, interface design, functions, project planning and the cost of free apps.

Volume 3 · Machines and Meaning

Lists, Boolean logic, hardware, files, requirements, machine learning, DNS, search ranking, control systems and copyright.

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Physical Computing with the Raspberry Pi 5 · AI Literacy for Secondary Students · and more.

Found an error, or want a lesson on something else?

Feedback is genuinely read. If a lesson did not land with your class, or you need a sub plan on a topic that is not here, get in touch through digitalvector.com.au and it will be considered for the next update.

EMERGENCY SUB PLANS PACK

Digital Technologies · Grades 7-10

Volume Four · ten lessons · no preparation · no computers required